User Study – ML2++ (Textual DSL)

Instructions: Fill Section A once. For Section B, answer for *each* use case you worked on. If you did multiple use cases, **duplicate Section B** per use case.

Participant ID: ____________________ **Use case(s):** ☐ U1 River Flow ☐ U2 Smart-Home Energy ☐ U3 Solar Energy

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Section A – Background (answer once)

This section helps us understand your experience level so we can analyze results fairly.

A1. Educational level *(highest completed)*

Tell us the highest level of education you have finished.

☐ High School ☐ Vocational/Technical ☐ Bachelor's ☐ Master's ☐ Doctorate ☐ Other: ____________________

A2. Primary role *(pick the closest)*

Choose the option that best matches your current job or main activity. If none fits, select "Other".

☐ IoT/CPS Developer ☐ MDE Practitioner ☐ Data Scientist / ML Engineer ☐ Embedded SW Engineer ☐ Educator / Researcher ☐ Software Architect / Full-stack ☐ System Integrator ☐ Early Adopter / Innovator ☐ Other: ____________________

A3. Self-rated expertise *(tick one per row)*

For each area, select your skill level. "IoT/CPS" = connected devices/systems. "MDE" = using models to guide or generate code. "ML/Data Science" = working with data and ML.

	Beginner	Intermediate	Advanced	Expert
IoT/CPS development	☐	☐	☐	☐
Model-Driven Engineering (MDE)	☐	☐	☐	☐
Machine Learning / Data Science	☐	☐	☐	☐

A4. Prior experience *(used DSLs/modeling tools/ML frameworks before?)*

Have you used similar tools (e.g., modeling editors, DSLs, ML libraries/frameworks)?

☐ Yes ☐ No

Section B – Task-Specific (duplicate this page *per use case*)

This B-section is for: ☐ U1 River Flow ☐ U2 Smart-Home Energy ☐ U3 Solar Energy

If you did more than one use case, copy this whole Section B for each one and fill it separately.

B1. Time & Effort

Please record your times honestly. If unsure, give your best estimate. "Extra learning time" = time reading docs/tutorials to figure things out.

B1. Time spent (minutes)

Minutes you spent defining/editing the DSL, generating code, reviewing outputs, and any extra learning time.

	ML2++ (Textual DSL)
Defining DSL + generate + review	40 min
Extra learning time	all time

B2. Statements written (DSL) *(exclude blanks/comments)*

Approximate number of DSL lines/statements you actually typed or edited.

	ML2++ (Textual DSL)
ML2++ DSL statements	159

B2. Task Completion & Ease of Use

Tell us if you finished the task and how easy/hard each step felt. Then write how many minutes you spent per step.

B3. Were you able to finish the use case?

"None" = could not complete; "Partial" = finished some parts; "Complete" = finished everything.

☐ None ☐ Partial ☒ Complete

B4. Ease by workflow step (with per-step time)

Rate difficulty (1=very easy, 5=very difficult) and write minutes spent. "Model configuration" = choosing algorithm and hyperparameters. "Model evaluation" = checking results with plots/metrics.

Step	1	2	3	4	5	**Time**
Whole workflow (end-to-end)	☐	☐	☒	☐	☐	10 min
Data preprocessing	☐	☐	☒	☐	☐	5 min
Model configuration (algorithm, hyperparameters)	☐	☐	☒	☐	☐	15 min
Model training	☐	☐	☒	☐	☐	5 min
Model evaluation (metrics/plots/inspection)	☐	☐	☒	☐	☐	5 min

Tip: use a simple stopwatch and pause for breaks. If unsure, estimate.

B5. Ease of ML2++ elements *(1=Very easy → 5=Very difficult)*

Rate how easy it was to work with each part of the language/editor.

Element	1	2	3	4
Data types & objects	☐	☐	☒	☐
Things & behaviours (statecharts)	☐	☐	☒	☐
Ports & properties	☐	☐	☐	☐
Model configuration options (algorithm/hyperparameters)	☐	☒	☐	☐
Configuration (instances/connectors)	☐	☒	☐	☐

B3. Learning & Usability

These questions focus on how quickly you learned the tool and how natural it felt.

B6. Learning curve *(1=Very easy → 5=Very difficult)*

How much effort did it take to learn enough to finish this use case?

☐1 ☒2 ☐3 ☐4 ☐5

B7. Intuitiveness *(1=Very intuitive → 5=Not intuitive)*

Did the tool feel natural/obvious to use, or did it feel confusing?

☐1 ☐2 ☐3 ☒4 ☐5

B8. Hardest part to learn *(briefly)*

Name the single most confusing or time-consuming thing to learn.

the sensor and server part

B4. Debugging & Error Handling

Tell us about problems you faced and how hard they were to fix.

B9. Error frequency *(0=None, 1=Few, 2=Moderate, 3=Many)*

How many problems blocked you (errors, failed runs, missing packages, wrong paths, etc.)?

☐0 ☐1 ☒2 ☐3

B10. Clarity of error messages *(1=Not clear → 5=Very clear)*

When something went wrong, did the messages help you understand the fix?

☒1 ☐2 ☐3 ☐4 ☐5

B11. Debugging effort *(1=Easy → 5=Hard)*

Overall, how hard was it to fix your issues?

☐1 ☐2 ☐3 ☒4 ☐5

B12. Debugging time (minutes) *(estimate)*

About how many minutes did you spend fixing problems (total for this use case)?

6 times

B5. Code Quality, Maintainability, Flexibility

These questions refer to the generated code you viewed and how easy it would be to change things later.

B13. Generated code quality *(1=Poor → 5=Excellent)*

How good is the generated code (readable, well-structured, correct)?

☐1 ☐2 ☐3 ☐4 ☐5

B14. Maintainability *(1=Very difficult → 5=Very easy)*

If you had to update/change your solution later (e.g., new dataset, new horizon), how easy would it be?

☐1 ☐2 ☐3 ☐4 ☐5

B15. Flexibility / Customisation *(1=None → 5=Very flexible)*

How easily can you change models, hyperparameters, thresholds, or add steps without workarounds?

☐1 ☐2 ☐3 ☐4 ☐5

Section C – Overall Assessment (ML2++ only)

Your overall opinion and which features helped you most.

C1. Overall satisfaction *(1=Very dissatisfied → 5=Very satisfied)*

Your overall feeling about using ML2++ for this use case.

☐1 ☐2 ☐3 ☐4 ☐5

C2. Would you use ML2++ for your next similar project?

If you had a similar project tomorrow, would you choose this approach? Tell us briefly why.

☐ Yes ☐ No ☐ Undecided *Why?* ____________________

C3. Feature effectiveness *(1=Ineffective → 5=Very effective)*

How useful were these features for you?

Feature	1	2	3	4	5
Time-series forecasting models	☐	☐	☐	☐	☐
Automatic preprocessing & plots	☐	☐	☐	☐	☐
Training configuration options	☐	☐	☐	☐	☐
Multi-day forecasting accuracy	☐	☐	☐	☐	☐

C4. Favourite feature: plots

What was the single most helpful feature?

C5. Least favourite feature: evaluation

Which feature felt least helpful or got in your way?

C6. Missing features you would like: separate training part

What do you wish the tool had (e.g., a plot, algorithm, wizard, or shortcut)?

Section D – Open Feedback

Tell us anything else that would help us improve.

D1. Barriers to adoption *(what would stop you from using ML2++ in production?)*

Examples: setup, missing plots, docs, integration, performance, governance.

being hard to write codes

D2. One improvement you would prioritize

If we could fix/add only one thing, what would help you the most?

Better setup for training and evaluation

Scale anchors used throughout for consistency:

- Ease / Learning: 1=Very easy, 2=Easy, 3=Moderate, 4=Difficult, 5=Very difficult.
- Intuitiveness: 1=Very intuitive, 2=Intuitive, 3=Neutral, 4=Somewhat confusing, 5=Not intuitive.
- Error clarity: 1=Not clear, 2=Somewhat clear, 3=Clear, 4=Very clear, 5=Extremely clear.
- Code quality: 1=Poor, 2=Fair, 3=Good, 4=Very good, 5=Excellent.
- Maintainability: 1=Very difficult, 2=Difficult, 3=Moderate, 4=Easy, 5=Very easy.
- Flexibility: 1=None, 2=Limited, 3=Some, 4=Flexible, 5=Very flexible.
- Feature effectiveness: 1=Ineffective, 2=Low, 3=Moderate, 4=Effective, 5=Very effective.